ENVS193DS Final

Callie Baker

Table of contents

GitHub repo 2

Set up 2

Problem 1. Research writing 3
- a. Transparent statistical methods . . . 3
- b. Suggestions for rewriting . . . 3
- c. Further analysis . . . 3

Problem 2. Data analysis 3
- a. Clean and wrangle . . . 4
- b. Response variable . . . 4
- c. Purpose of study . . . 4
- d. Table of models . . . 4
- e. Run the models . . . 4
- f. Select the best model . . . 4
- g. Check the diagnostics . . . 4
- h. Visualize the model predictions . . . 4
- i. Write the caption for your figure . . . 4
- j. Calculate model predictions . . . 4
- k. Interpret your results . . . 4

Problem 3. Affective and exploratory visualizations 4
- a. Comparing visualizations . . . 4
- b. Designing an analysis . . . 5
- c. Check any assumptions and run your analysis . . . 5
  - Think about your data . . . 5
  - Write your model . . . 5
  - Look at the diagnostics . . . 6
  - Look at the model coefficients . . . 7

Calculate an effect size 7
Looking at the model predictions 8
d. Create a visualization 9
Visualizing model predictions 9
Describing the model 11
e. Write a caption 11
f. Write about your results 11
g. Sharing your affective visualization 11

GitHub repo

Access my repo here.

Set up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization
library(janitor) # cleaning data frames
library(DHARMa) # for GLM diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
library(gtsummary) # for GLM

# read in data
nests <- read_csv(here("data", "nest_data_final.csv"))
# read in personal data to object from csv
my_data <- read_csv(here("data", "envs193ds-personal-data.csv"))
```

```
# setting up personal data for problem 3

# cleaned column names
my_data <- my_data |>
  clean_names()

# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
```

```
    workout_type == "Run" ~ 1, # run = 1
    .default = 0))             # all other workout types = 0
```

Problem 1. Research writing

a. Transparent statistical methods

Part 1: The statistical test used in Part 1 was a generalized linear model (logistic regression) with a binomial distribution. The response variable is a binary outcome to reflect the probability that the American Avocet uses the mircohabit (1) or doesn't (0). The predictor is a continuous variable: distance to water edge (UNITS).

Part 2: The statistical test used in Part 2 was a chi-square test of independence which is used to determine whether there is a relationship between two categorical variables. The categorical variables are bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat). The test compares observed counts to expected counts.

b. Suggestions for rewriting

Part 1: American Avocets were more likely to use microhabitats that were closer to the water's edge, and this relationship was strong enough that is unlikely to be the result of random chance alone.

Part 2: The three shorebird species did not use the five habitat types in equal proportions: each species showed a distinct pattern of habitat preference, and these differences across species are unlikely to reflect chance.

c. Further analysis

Tidal inundation frequency could influence American Avocet microhabitat use because Avocets forage in shallow flooded areas so microhabitats that are too rarely or too frequently inundated may not provided the water depth conditions needed for successful foraging. Tidal inundation frequency is a continuous variable, measured as the proportion of time a microhabitat is flooded per tidal cycle; this variable could be recorded using tidal monitoring stations.

Problem 2. Data analysis

*DO NOT include any code or output from your data exploration

a. Clean and wrangle

```
# cleaned data
nests_clean <- nests |>
  clean_names()
```

b. Response variable

c. Purpose of study

d. Table of models

e. Run the models

f. Select the best model

g. Check the diagnostics

h. Visualize the model predictions

i. Write the caption for your figure

j. Calculate model predictions

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

-
-
-
-

b. Designing an analysis

I used a binomial GLM to ask whether sleep duration the previous night (continuous, measured in hours) predicts the probability that my workout was a run (binary, 1 = run and 0 = all other workout types). This analysis was appropriate because the response variable is binary, so a standardized linear model can not be used as the assumption of normally distributed errors is not met. Therefore, I used a GLM with a binomial error distribution because this test is specifically designed for response variables that have outcomes of 0/1. Sleep is a continuous predictor variable, so it works as the explanatory variable in my logistic regression.

c. Check any assumptions and run your analysis

Think about your data

The assumptions for a binomial GLM are met. First, each workout entry represents a separate day, so observations are independent. Second, there are binary outcomes, seen through the response variable run having 1 (a run occurred) and 0 (a run did not occur) outcomes. Finally, there is a linear relationship between transformed expected response in terms of link function and explanatory variables (WHAT DOES THIS MEAN – REPHRASE).

Write your model

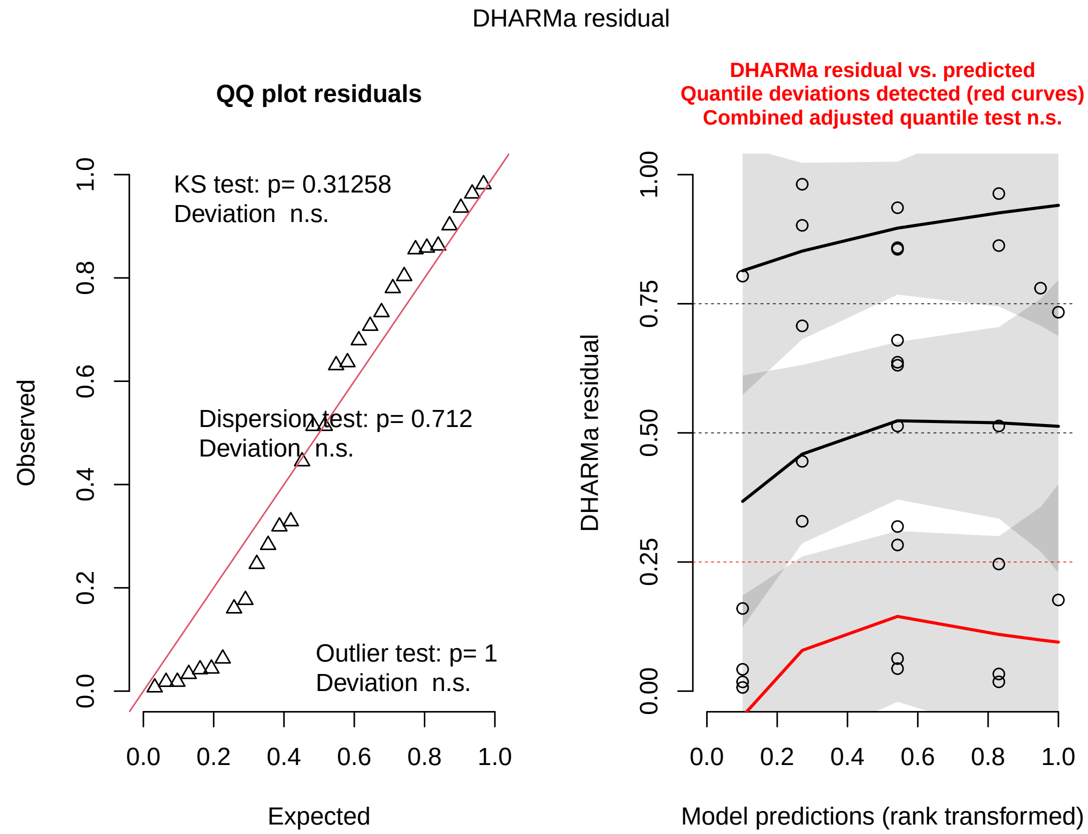
In words: Sleep duration the previous night predicts the occurrence of going on a run (yes = 1, no = 0).

In code:

```
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
  # formula: binary run outcome ~ sleep hours
  is_run ~ sleep_hours,
  # data: my_data data frame
  data = my_data,
  # binomial error distribution for binary response
  family = binomial)
```

Look at the diagnostics

```
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(run_mod))
```



Information from left plot (QQ plot of residuals):

- **KS test**: Do my residuals follow a uniform distribution? → Generally yes
- **Dispersion test**: Are your residuals more dispersed than expected from a uniform distribution? → No
- **Outlier test**: Are there outliers in your data set that could be influencing model coefficients or predictions? → No

Information from right plot (residuals vs predicted values):

In visually assessing for randomly arranged residuals across the range of predicted values, DHARMa diagnostics flag issues in the 25th quartile. This is likely due to the small sample size, not because the model is wrong. Therefore, it it reasonable to say that residuals are randomly arranged across the range of predicted values.

Look at the model coefficients

```r
# displayed model summary
summary(run_mod)
```

```

Call:
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)  
(Intercept)  -8.7011     3.4964  -2.489   0.0128 *
sleep_hours   1.0873     0.4712   2.307   0.0210 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 36.652  on 29  degrees of freedom
Residual deviance: 28.046  on 28  degrees of freedom
AIC: 32.046

Number of Fisher Scoring iterations: 5
```

Note: - all estimates are log odds, so calculate the probability of a run when sleep is 0 below using inverse logit

Calculate an effect size

```r
# calculated inverse logit
plogis(-8.7011)
```

```
[1] 0.000166375
```

Interpretation:

Characteristic	OR	95% CI	p-value
sleep_hours	2.97	1.37, 9.40	0.021

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

- inverse logit: 0.00016
- the probability of going on a run if sleep duration is 0 hours is 0.00016

```
# calculated odds ratio
gtsummary::tbl_regression(run_mod,
                          exponentiate = TRUE)
```

Interpretation:

- results table rendered at the top of this page
- odds ratio: 2.97
- 2.97 > 1 so the probability of going on a run increases by a factor of 2.97 with every 1 hour increase in sleep (95% CI: [1.37, 9.40])

Looking at the model predictions

- using probabilities to determine chance of getting certain outcome
- line represents the probability of 1 (going on a run) actually occurring

```
# what is the probability of going on a run after sleeping for 10 hours?
ggpredict(run_mod,
          terms = "sleep_hours [10]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
         10 |      0.90 | 0.39, 0.99
```

```
# what is the probability of going on a run after sleeping for 4 hours?
ggpredict(run_mod,
          terms = "sleep_hours [4]")
```

```
# Predicted probabilities of is_run

sleep_hours | Predicted |     95% CI
------------------------------------
          4 |      0.01 | 0.00, 0.25
```

Note:

- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).

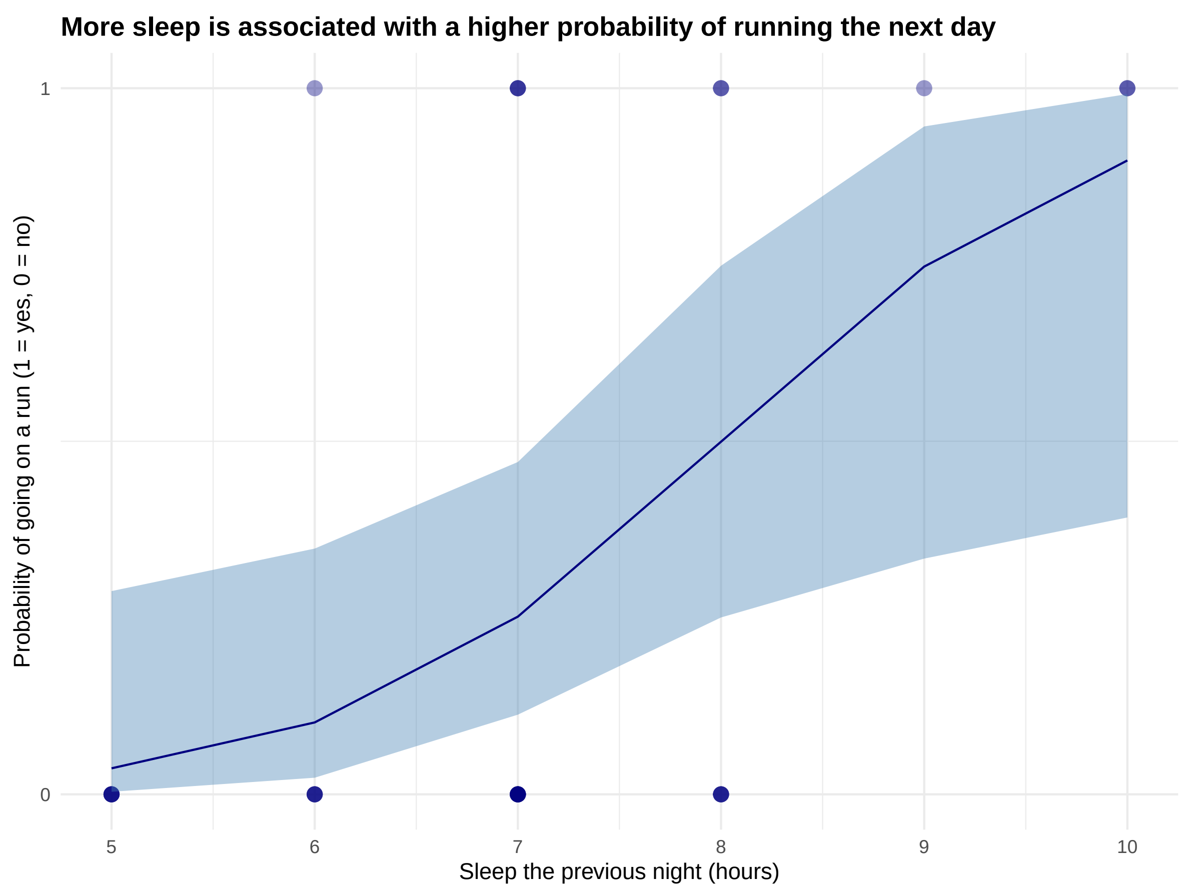
d. Create a visualization

Visualizing model predictions

```r
# generated model predictions across range of sleep values in data set
mod_preds <- ggpredict(run_mod,
                       terms = "sleep_hours [5:10 by = 1]")

# base layer: ggplot with raw data
ggplot(my_data,              # data frame: my_data
       aes(x = sleep_hours, # x-axis: sleep duration
           y = is_run)) +    # y-axis: probability of running (0/1)
  # first layer: raw observations
  geom_point(size = 3,              # made points larger
             alpha = 0.4,           # made points more transparent
             color = "navyblue") + # changed points to navy blue
  # second layer: confidence interval ribbon from model predictions
  geom_ribbon(data = mod_preds, # data frame: model predictions
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              fill = "steelblue", # changed ribbon color to steel blue
              alpha = 0.4) + # made ribbon more transparent
  # third layer: logistic curve from model predictions
  geom_line(data = mod_preds,
            aes(x = x,
                y = predicted),
```

```
            color = "navyblue") +  # changed line color to navy
  # constrained y-axis to 0 and 1
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1)) +
  # relabeled x- and y-axis
  # added title
  labs(title =
"More sleep is associated with a higher probability of running the next day",
       x     = "Sleep the previous night (hours)",
       y     = "Probability of going on a run (1 = yes, 0 = no)") +
  # changed theme from default
  theme_minimal() +
  # bolded title
  theme(plot.title = element_text(face = "bold"))
```



Describing the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(>lzl)	
(Intercept)	-8.701	3.496	-2.489	0.0128	*
sleep_hours	1.087	0.471	2.307	0.0210	*

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom

e. Write a caption

Figure 1. Longer sleep duration is associated with a higher probability of going on a run the next day. Blue circles represent individual workout observations (0 = not a run, 1 = run). The navy blue line represents model predicted probability of running across the range of observed sleep values (5-10 hours), and the shaded blue ribbon represents the 95% confidence interval around model predictions. Data collected from personal fitness tracking records between 2026-04-14 and 2025-05-26.

f. Write about your results

I tended to go on runs more often on days following nights with more sleep. For every 1 hour increase in sleep the previous night, the odds of going on a run increased by a factor of 2.97 (95% CI: [1.37, 9.40], p = 0.021, $\alpha = 0.05$). After sleeping for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]), compared to a probability of 0.01 (95% CI [0.00, 0.25]) after sleeping for only 4 hours. This suggests that sleep duration has an influence on my likelihood of running.

g. Sharing your affective visualization

Presented visualization in Week 10 Workshop.